

Polynomial Linear Regression

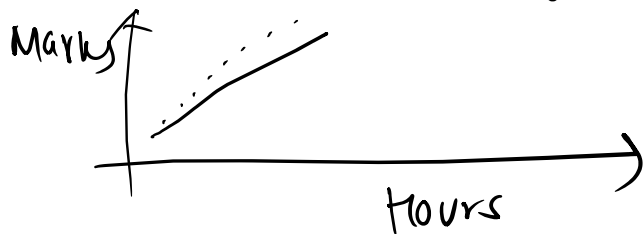
Wednesday, July 22, 2026 8:02 PM

Polynomial Regression is a machine learning algorithm used when the relationship between the input variable (X) and output variable (Y) is **curved (non-linear)** instead of a straight line.

Need of polynomial regression:-

Hours	Marks	[Linear relation]
1	20	}
2	40	
3	60	
4	80	

A straight line fits very well.



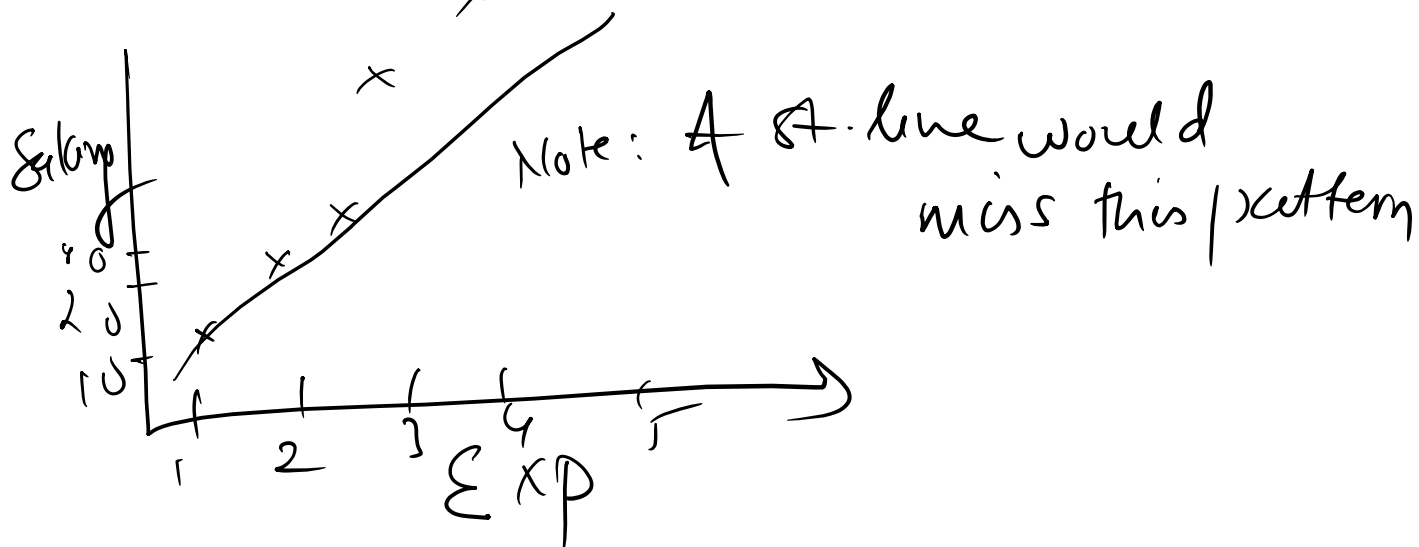
Linear regression eqⁿ: $y = b_0 + b_1 x$

Non-Linear Relationship

Exp	Salary
1	30k
2	40k
3	60k
4	95k
5	150k

5 1 150k]

Growth is not constant.



Idea behind Polynomial Regression:-

Instead of fitting

Expand

$$y = b_0 + b_1 x \longrightarrow [y = b_0 + b_1 x + b_2 x^2 + b_3 x^3 + \dots]$$

we fit

$$y = b_0 + b_1 x + b_2 x^2 + b_3 x^3 + \dots + b_n x^n$$

features;

$$x, x^2, x^3, x^4$$

Then we use Linear regression in these transformed features

Degree of a polynomial

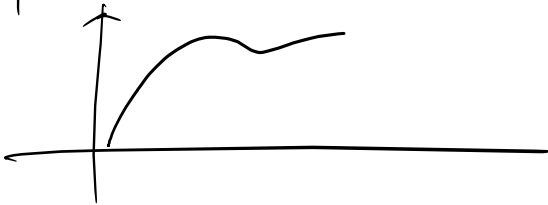
Degree - 1

$$y = b_0 + b_1x \quad [\text{S+ line}]$$

Degree - 2

$$y = b_0 + b_1x + b_2x^2$$

produces one curve (parabolic)



Degree - 3

$$y = b_0 + b_1x + b_2x^2 + b_3x^3$$

[S-shaped]

Working Example :-

Suppose X

1
2
3
4

Using degree = 3

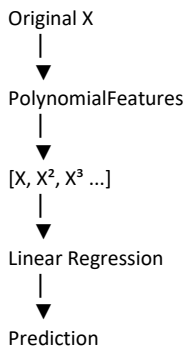
Using degree = 3

$$y = b_0 + b_1x + b_2x^2 + b_3x^3$$

x	x^2	x^3
1	1	1
2	4	8
3	9	27
4	16	64

$$y = b_0 + b_1(x) + b_2(x^2) + b_3(x^3)$$

Algorithm itself is still linear regression.



Advantages :-

- 1) capture curved relationships
- 2) more accurate than Linear regression when true relationship is non-linear.
- 3) faster to train (relatively)

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Disadvantages

- 1) Choosing a higher degree can cause overfitting. Training acc $\uparrow$ Testing acc $\downarrow$
- 2) Higher-degree models can be sensitive to noise.